A Few Notes on Simple Harmonic Oscillators

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1 Introduction

This story begins somewhere around 1658 when Robert Hooke [5] began experimenting with springs and masses. By 1660 Hooke had made two significant steps, namely the use of a balance controlled by a spiral spring and an improved escapement called the anchor escapement. In 1660 he discovered an instance of what we now call Hooke's Law while working on designs for the balance springs of clocks. However he only announced the general law of elasticity in his lecture Of Spring given in 1678 [4].

Interestingly, it was the addition of the balance spring to the balance wheel around 1658 by Robert Hooke (and Christiaan Huygens [14]) that greatly increased the accuracy of portable timepieces, transforming early pocket watches from expensive novelties to useful timekeepers.

These notes are organized (I hope) as follows: Section 2 looks at Hooke's Law in some detail. Section 3.1 considers the simple pendulum and Section 3.2 considers the RLC circuits as examples of other simple harmonic oscillators. Section 4 looks at the Quantum Harmonic Oscillator. Section 5 outlines how quantum fields fit into all of this. Finally, 6 offers a few observations and conclusions.

2 Hooke's Law

Hooke's law [16] is a law of physics that states that the force (I'll just call it F) needed to extend or compress a spring by some distance (x) scales linearly with respect to that distance, that is, $F_s = -kx$, where k is a constant factor characteristic of the spring (i.e., its stiffness), and x is small compared to the total possible deformation of the spring. The law is named after 17th-century British physicist Robert Hooke [5]. He first stated the law in 1676 as a Latin anagram, and he published the solution of his anagram in 1678 as: ut tensio, sic vis ("as the extension, so the force" or "the extension is proportional to the force") [4].

2.1 The Mathematics of Hooke's Law

Hooke's Law is most frequently stated as $F \propto -x$, that is, the force F is proportional to the displacement x. However, more frequently this relationship is changed into an equality by the use of a spring constant k so that we have[1]

$$F = -kx \tag{1}$$

We also know that Newton's Second Law of Motion [17] tells us that

$$F = ma \tag{2}$$

[1] It seems to be convention to use x and $x(t)$ and a and $a(t)$ interchangeably.

Now, if we set Equation 1 equal to Equation 2 we see that

$$
\begin{aligned}
ma &= -kx && \text{\# Newton's force equals Hooke's force} \\
\Rightarrow \quad a &= -\frac{k}{m}x && \text{\# solve for } a\text{, noting that } a \text{ and } x \text{ are functions of time} \\
\Rightarrow \quad \frac{d^2x}{dt^2} &= -\frac{k}{m}x && \text{\#} \; a = \frac{d}{dt}(v) = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2}
\end{aligned}
$$

This situation is shown in Figure 1.

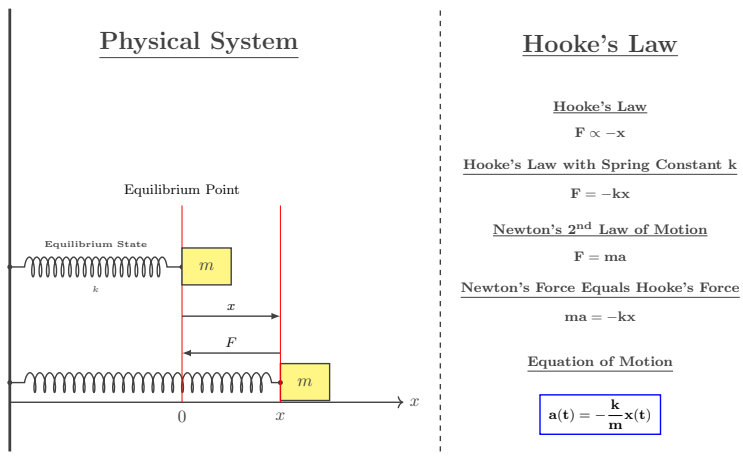

Figure 1: Spring and Mass System with Spring Constant k and Mass m

So now we know that

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x \tag{3}$$

The next question is how do we solve the second order ordinary differential equation (Equation 3) for $x(t)$? Well, if we imagine the displacement ($x(t)$) plotted on the y axis against t (plotted on the x axis) we see the familiar pattern shown in Figure 2.

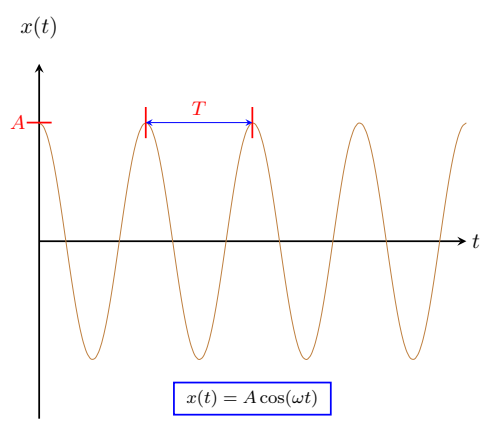

Figure 2: Simple Harmonic Motion Displacement Function

Here we can think of the spring being stretched to its maximum displacement (A) and then let go at time $t = 0$. The mass (m) will continue oscillating since in the Hooke's Law setup the only force acting on the mass is the force due to the spring, that is, $-kx$. This force is frequently referred to as the *restoring* force and is shown in Figure 1.

We can check to see if our guess, $x(t) = A\cos(\omega t)$, is really a solution to Equation 3 as follows:

$$
\begin{aligned}
\frac{d^2x}{dt^2} &= -\frac{k}{m}x && \text{\# Equation 3} \\
&\Rightarrow \frac{d^2}{dt^2}\left[A\cos(\omega t)\right] = -\frac{k}{m}x && \text{\# guess that } x(t) = A\cos(\omega t) \\
&\Rightarrow \frac{d}{dt}\left[-\omega A\sin(\omega t)\right] = -\frac{k}{m}x && \text{\# } \frac{d}{dt}\cos(u) = -\sin(u)\frac{du}{dt} \text{ with } u = \omega t \\
&\Rightarrow -\omega^2 A\cos(\omega t) = -\frac{k}{m}x && \text{\# } \frac{d}{dt}\sin(u) = \cos(u)\frac{du}{dt} \text{ with } u = \omega t \\
&\Rightarrow \omega^2 A\cos(\omega t) = \frac{k}{m}x && \text{\# cancel minus} \\
&\Rightarrow \omega^2 x = \frac{k}{m}x && \text{\# } x(t) = A\cos(\omega t) \\
&\Rightarrow \omega^2 = \frac{k}{m} && \text{\# cancel } x \\
&\Rightarrow \omega = \sqrt{\frac{k}{m}} && \text{\# take the square root of both sides}
\end{aligned}
$$

So $x(t) = A\cos(\omega t)$ is indeed a solution to Equation 3 in the case that

$$\omega = \sqrt{\frac{k}{m}} \tag{4}$$

Here A is the maximum displacement (or amplitude) and ω is the angular frequency (in radians per second). We also know that the definition of angular frequency is

$$\omega = \frac{2\pi}{T} = 2\pi f \tag{5}$$

where T is the period (time for one cycle in seconds) and f is the frequency in cycles/second [12]. Note that $f = \frac{1}{T}$ so f has units of seconds^{-1}.

Give this information we can write the displacement function $x(t)$ as in terms of the frequency f as follows:

$$x(t) = A\cos(2\pi f t)$$

We can also put Equations 4 and 5 together to find

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

and

$$T = 2\pi\sqrt{\frac{m}{k}}$$

2.2 Total Energy of a Simple Harmonic Oscillator

We know that the total energy of an oscillator such as the spring and mass system we looked at above is

$$E_{\text{total}} = \text{KE} + \text{PE}$$

All good, but what is the kinetic energy (KE) and potential energy (PE) of such as system?

Well, we know that [6]

$$E_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

so that

$$\begin{aligned}
E_{\text{total}} &= \frac{1}{2}mv^2 + \frac{1}{2}kx^2 && \#\ E_{\text{total}} = \text{KE} + \text{PE} \\
&= \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + \frac{1}{2}kx^2 && \#\ v = \frac{dx}{dt} \\
&= \frac{1}{2}m\left(\frac{d}{dt}\left[A\cos(\omega t)\right]\right)^2 + \frac{1}{2}kx^2 && \#\ x(t) = A\cos(\omega t) \\
&= \frac{1}{2}m\left(-\omega A\sin(\omega t)\right)^2 + \frac{1}{2}kx^2 && \#\ \frac{d}{dt}\cos(u) = -\sin(u)\frac{du}{dt} \text{ with } u = \omega t \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}kx^2 && \#\ \left(-\omega A\sin(\omega t)\right)^2 = \omega^2 A^2\sin^2(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}k\left(A\cos(\omega t)\right)^2 && \#\ x(t) = A\cos(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}kA^2\cos^2(\omega t) && \#\ \left(A\cos(\omega t)\right)^2 = A^2\cos^2(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}m\omega^2 A^2\cos^2(\omega t) && \#\ \omega = \sqrt{\frac{k}{m}} \Rightarrow k = m\omega^2 \text{ (Equation 4)} \\
&= \frac{1}{2}m\omega^2 A^2\left(\sin^2(\omega t) + \cos^2(\omega t)\right) && \#\ \text{factor out } \frac{1}{2}m\omega^2 A^2 \\
&= \frac{1}{2}m\omega^2 A^2 && \#\ \sin^2(\omega t) + \cos^2(\omega t) = 1 \\
&= \frac{1}{2}m(2\pi f)^2 A^2 && \#\ \omega = 2\pi f \\
&= \frac{1}{2}4\pi^2 f^2 A^2 m && \#\ \text{expand } (2\pi f)^2 \text{ and rearrange} \\
&= 2\pi^2 f^2 A^2 m && \#\ E_{\text{total}} = 2\pi^2 f^2 A^2 m \\
&\Rightarrow E_{\text{total}} \propto f^2 A^2 && \#\ E_{\text{total}} \text{ is proportional to } f^2 A^2
\end{aligned}$$

So now we know that the oscillator has $E_{\text{total}} = 2\pi^2 f^2 A^2 m$, or said another way $E_{\text{total}} \propto f^2 A^2$. This result will become useful when we consider the quantum harmonic oscillator.

3 Other Simple Harmonic Oscillators

3.1 The Simple Pendulum

A pendulum is a weight (that is, a mass m in a gravitational field) suspended from a pivot so that it can swing freely [18]. When a pendulum is displaced sideways from its resting, equilibrium position, it is subject to a restoring force due to gravity that will accelerate it back toward the equilibrium position. When released, the restoring force acting on the pendulum's mass causes it to oscillate about the equilibrium position, swinging back and forth. The time for one complete cycle, a left swing and a right swing, is called the period (T). The period depends on the length of the pendulum and also to a slight degree on the amplitude (A), the width of the pendulum's swing. The free body diagram [15] for the simple pendulum is shown in Figure 3.

From the first scientific investigations of the pendulum around 1602 by Galileo Galilei, the regular motion of pendulums was used for timekeeping, and was the world's most accurate timekeeping technology until the 1930 [11]. The pendulum clock invented by Christiaan Huygens in 1658 became the world's standard timekeeper [14], used in homes and offices for 270 years, and achieved accuracy of about one second per year before it was superseded as a time standard by the quartz clock in the 1930s. Pendulums are also used in scientific instruments such as accelerometers and seismometers. Historically they were used as gravimeters to measure the acceleration of gravity in geo-physical surveys, and even as a standard of length. The word "pendulum" is new Latin, from the Latin *pendulus* meaning 'hanging' [20].

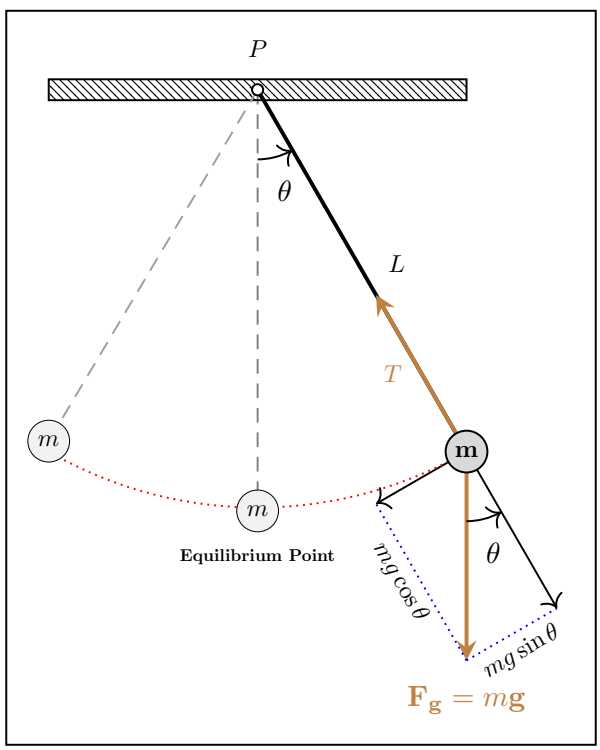

Figure 3: Free Body Diagram for a Simple Pendulum

The simple pendulum setup has a point mass m (the bob) on a string (or rod) of length L. Here the mass of the string is assumed to be negligible as compared to the mass of the bob. The pendulum pivots around a frictionless point P on the support. The only forces acting on the bob are the force of gravity (i.e., the weight of the bob) and tension from the string (T). The forces on the bob result in a net force of $-mg\sin\theta$ toward the equilibrium point. This force is called a *restoring force* since it points in the direction of the equilibrium point. The simple pendulum setup is shown in Figure 3.

3.1.1 Applying Newton's Laws to the Simple Pendulum

The pendulum is governed by Newton's laws applied to rotational motion, namely

$$\tau = I\alpha \tag{6}$$

where τ is the torque, I is the moment of inertia, and α is the angular acceleration [9].

Torque is interesting. It is a measure of rotational force, that is, it is a measure of twist. The moment of inertia of an object, also known as its rotational inertia, is a measure of how difficult it is to change the rotation of that object. It is an analogue of mass: mass is to force as rotational inertia is to torque. Angular acceleration is simply the change in velocity of the angle θ of the pendulum and is typically denoted[2] by either $\ddot{\theta}(t)$ or $\frac{d^2\theta}{dt^2}$.

A torque on a lever can be expressed as $\boldsymbol{\tau} = \mathbf{F} \times \mathbf{r}$, a force $\mathbf{F}$ multiplied by a distance from the pivot $\mathbf{r}$. In this case (Figure 3), $\mathbf{r} = L$ and $\mathbf{F} = m\mathbf{g}\sin\theta$, the force due to gravity that is perpendicular to the string. This force can be found by splitting the total gravitational force into its parallel and perpendicular parts through vector algebra (BTW, this is one of the reasons to like Lagrangian mechanics over Newtonian mechanics; Lagrangian mechanics uses (mostly) scalars whereas Newtonian mechanics uses vectors). This operation seems to be common and physics. In the case shown in Figure 3 we find that $\mathbf{F} = m\mathbf{g}\sin\theta$, the perpendicular force is the product of the total gravitational force $m\mathbf{g}$ and the scale factor $\sin\theta$.

Looking at the right hand side of Equation 6 we see we need to define I, the moment of inertia of the pendulum. In general the moment of inertia is found for any given object by $I = \sum mr^2$, where the sum is over each point in the object. Since the mass in our pendulum is concentrated in a single point at the bob, the sum is over a single term, the bob. Therefore

$$I = mL^2 \tag{7}$$

If we combine Equations 6 and 7 we see that

$$\begin{aligned}
\tau &= I\alpha && \text{\# definition of torque } \tau \text{ (Equation 6)} \\
&= mL^2\alpha && \text{\# definition of moment of inertia } I \text{ (Equation 7)} \\
&\Rightarrow -\mathrm{F}L = mL^2\alpha && \text{\# } \tau = \mathrm{F}L \\
&\Rightarrow -L\,mg\sin\,\theta = mL^2\alpha && \text{\# } \mathrm{F} = mg\,\sin\,\theta \\
&\Rightarrow -\,g\sin\,\theta = L\alpha && \text{\# cancel } L \\
&\Rightarrow \alpha = -\frac{g}{L}\sin\,\theta && \text{\# solve for } \alpha
\end{aligned}$$

Notice that the torque was made negative. This is by convention, since a negative torque indicates force in the clockwise direction, and the force of gravity perpendicular to the pendulum rod is clockwise at the initial state of the pendulum.

So now we know that

$$\alpha = -\frac{g}{L}\,\sin\,\theta \tag{8}$$

Now, if we let $\omega = \sqrt{\frac{g}{L}}$ and recall that $\alpha = \frac{d^2\theta}{dt^2}$ we see that

[2] Here again θ and $\theta(t)$ are used interchangeably.

$$
\begin{aligned}
\alpha &= -\frac{g}{L}\sin\theta && \text{\# Equation 8} \\
&= -\omega^2 \sin\theta && \text{\# } \omega = \sqrt{\frac{g}{L}} \\
\Rightarrow \quad & \frac{d^2\theta}{dt^2} = -\omega^2 \sin\theta && \text{\# } \alpha = \frac{d^2\theta}{dt^2}
\end{aligned}
$$

Now we know that

$$\frac{d^2\theta}{dt^2} = -\omega^2 \sin\theta \tag{9}$$

But here's an interesting problem: Unlike the equation of motion for the spring and mass problem (Equation 3), we can't solve the second order differential equation of motion for the simple pendulum case (Equation 9). However, we can solve a different but related problem which is called the "Small-Angle Pendulum Problem".

3.1.2 The Small-Angle Pendulum Problem

Interestingly, it turns out that for small θ (say, $\theta \leq 10°$) we see that $\sin\theta \approx \theta$. Said another way

$$\lim_{x\to 0} \frac{\sin x}{x} = 1$$

Using this constraint and Equation 9 we can write

$$\frac{d^2\theta}{dt^2} \approx -\omega^2\theta \tag{10}$$

This is a much simpler differential equation, which we know how to solve[3]. In order to solve this differential equation, we first notice that that the second derivative $\frac{d^2\theta}{dt^2}$ is the function itself, scaled by the constant $-\omega^2$. This form is known as a second order linear homogeneous differential equation [10] and we can guess the function that exhibits this behavior. It's the exponential function:

$$\frac{d}{dx}e^x = e^x$$

So if we set $\theta = e^{i\omega t}$ and recalling that $\frac{d}{dt}e^u = e^u\frac{du}{dt}$ we can see that

$$
\begin{aligned}
\theta &= e^{i\omega t} && \text{\# definition of } \theta \\
\frac{d\theta}{dt} &= i\omega e^{i\omega t} && \text{\# } \theta = e^{i\omega t} \text{ and } \frac{d}{dt}e^u = e^u\frac{du}{dt} \text{ with } u = i\omega t \\
\frac{d^2\theta}{dt^2} &= \frac{d}{dt}\left[i\omega e^{i\omega t}\right] && \text{\# second derivative is the derivative of the first derivative} \\
&= i^2\omega^2 e^{i\omega t} && \text{\# second derivative} \\
&= -\omega^2 e^{i\omega t} && \text{\# } i^2 = -1
\end{aligned}
$$

[3]Notice that this equation has the same form as the equation of motion for the spring and mass problem, namely $\frac{d^2x}{dt^2} = -\omega^2 x$.

So now we know that $\theta_1 = C_1 e^{i\omega t}$ is a solution to Equation 10, where $C_1 \in \mathbb{C}$. Notice also that $\theta_2 = C_2 e^{-i\omega t}$ is also a solution with $C_2 \in \mathbb{C}$. Here the (constant) coefficients C_1 and C_2 are preserved through differentiation and are essentially the integration constants. These constants represent the infinite number of solutions to the differential equation Equation 10, where are determined by the initial characteristics of the pendulum. The sum of these solutions is a more general solution, since either of the two constants be be set to θ to obtain the original solutions:

$$\theta = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \tag{11}$$

Taking the derivative of Equation 11 we get

$$\frac{d\theta}{dt} = C_1 i\omega e^{i\omega t} - C_2 i\omega e^{-i\omega t} \tag{12}$$

Taking the derivative one more time we see that

$$\begin{aligned}
\frac{d^2\theta}{dt^2} &= \frac{d}{dt}\Big[C_1 i\omega e^{i\omega t} - C_2 i\omega e^{-i\omega t}\Big] && \text{\# Equation 12} \\
&= C_1 i\omega e^{i\omega t}(i\omega) - C_2 i\omega e^{-i\omega t}(-i\omega) && \text{\# } \frac{d}{dt}e^u = e^u\frac{du}{dt} \text{ with } u = i\omega t \text{ and } i^2 = -1 \\
&= i^2 C_1 \omega^2 e^{i\omega t} + i^2 C_2 \omega^2 e^{-i\omega t} && \text{\# } (-1)*(-1) = +1\text{, collect terms} \\
&= -C_1\omega^2 e^{i\omega t} - C_2\omega^2 e^{-i\omega t} && \text{\# } i^2 = -1 \\
&= -\omega^2(C_1 e^{i\omega t} + C_2 e^{-i\omega t}) && \text{\# factor out } -\omega^2 \\
&= -\omega^2\theta && \text{\# } \theta = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \text{ (Equation 11)} \\
\Rightarrow \frac{d^2\theta}{dt^2} &= -\omega^2\theta && \text{\# Equation 10}
\end{aligned}$$

This result tells us that Equation 10 encompasses all of the possible solutions to the small-angle pendulum problem. All good, but one question we might ask is what happens when we apply Euler's formula [3] to Equation 11? Recall that Euler's Formula states that $e^{ix} = \cos x + i\sin x$ so we see that

$$\begin{aligned}
\theta &= C_1 e^{i\omega t} + C_2 e^{-i\omega t} && \text{\# Equation 11} \\
&= C_1\cos(\omega t) + C_1 i\sin(\omega t) + C_2 e^{-i\omega t} && \text{\# } C_1 e^{i\omega t} = C_1\cos(\omega t) + C_1 i\sin(\omega t) \\
&= C_1\cos(\omega t) + C_1 i\sin(\omega t) + C_2\cos(-\omega t) + C_2 i\sin(-\omega t) && \text{\# } C_2 e^{-i\omega t} = C_2\cos(-\omega t) + C_2 i\sin(-\omega t) \\
&= C_1\cos(\omega t) + C_1 i\sin(\omega t) + C_2\cos(\omega t) - C_2 i\sin(\omega t) && \text{\# } \sin(-\theta) = -\sin\theta \text{ and } \cos(-\theta) = \cos\theta \\
&= (C_1 + C_2)\cos(\omega t) + i(C_1 - C_2)\sin(\omega t) && \text{\# group terms, rearrange} \\
&= A\cos(\omega t) + B\sin(\omega t) && \text{\# } A = C_1 + C_2 \text{ and } B = i(C_1 - C_2)
\end{aligned}$$

So now we can write $\theta = A\cos(\omega t) + B\sin(\omega t)$ where $A, B \in \mathbb{C}$ are constants defined by the initial conditions of the pendulum.

Since θ is a linear combination of periodic functions it is also periodic [2]. We can also note here that adding two periodic functions with the same period results in another periodic function with that same period [2]. Since the periods of both $\cos(\omega t)$ and $\sin(\omega t)$ are $\frac{2\pi}{\omega}$ we know that the period of the pendulum, recalling that we defined $\omega = \sqrt{\frac{g}{L}}$ and applying the small-angle approximation, is

$$T \approx \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$$

An interesting outcome here is that this approximation of the period does not rely on the maximum angular displacement of the pendulum, which is in accordance with Galileo's observations as described in [1].

Summary: It turns out that the small-angle approximation for the pendulum period is fairly good and many papers study the error in this approximation. For example, when $\theta < \frac{\pi}{4}$ the relative (or approximation) error[4] R is less than 3.84%. However, even this small inaccuracy can add up: after about 13 full oscillations of a pendulum with $\omega = \pi$ and $\theta_{\max} = \frac{\pi}{4}$, the approximation is out of phase by half a period with the exact pendulum. This means that after 13 oscillations, when the approximate pendulum is at equilibrium, the exact pendulum is maximally displaced, and vice-versa [7].

Finally, note that pendulums with small angles are not uncommon. For example, many clocks have long, thin pendulums: essentially large L and small $\theta_{\max}$. In practice these two constants are used produce clocks with long, accurate periods [19].

3.1.3 Large-Angle Pendulums

TBD

3.2 The RLC Circuit Oscillator

TBD [8]

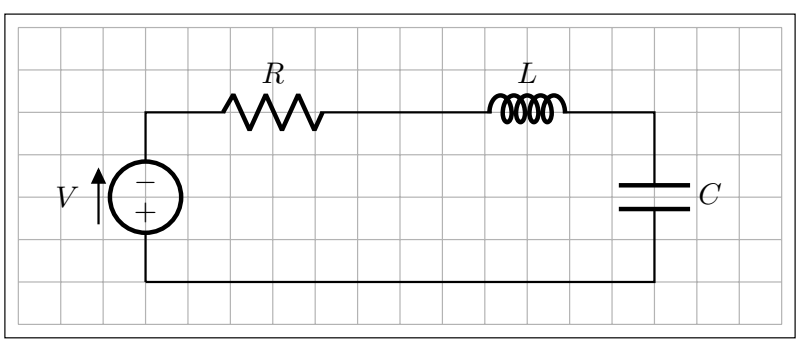

Figure 4: Series RLC Circuit

[4]The Relative Error R computed as follows: $R = 100 \cdot \frac{T_{\text{exact}} - T_{\text{approx}}}{T_{\text{exact}}}$ [13].

4 The Quantum Harmonic Oscillator

5 Quantum Fields

6 Conclusions

Acknowledgements

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Figures

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